

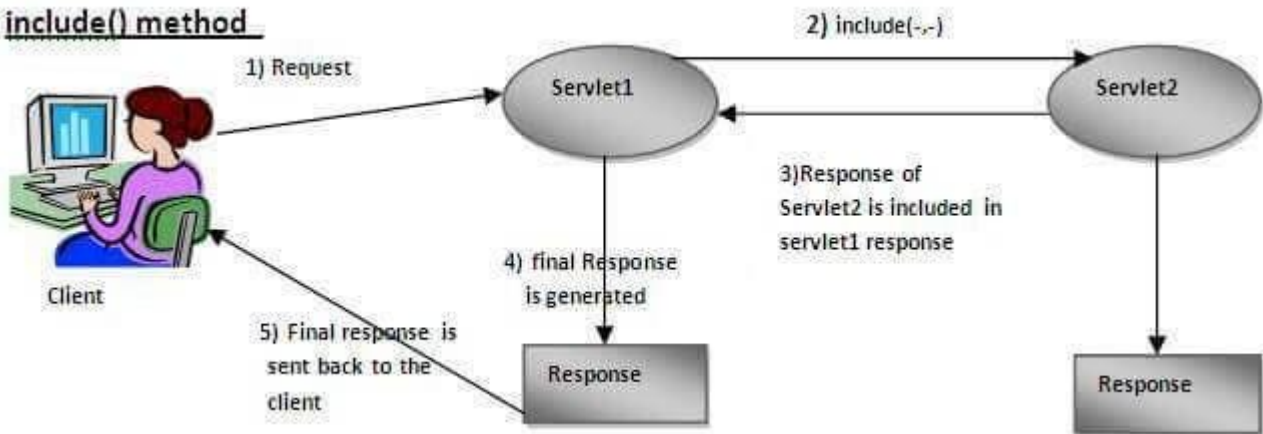
Servlet Collaboration

The RequestDispatcher interface provides the facility of dispatching the request to another resource it may be html, servlet or jsp. This interface can also be used to include the content of another resource also. It is one of the way of servlet collaboration.

There are two methods defined in the RequestDispatcher interface

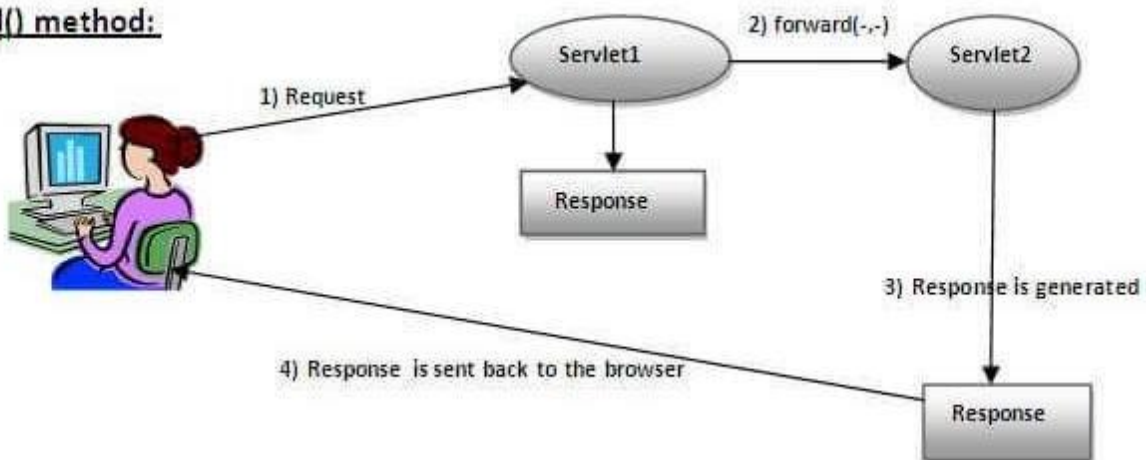
1. **public void forward(ServletRequest request,ServletResponse response)throws ServletException,java.io.IOException:**Forwards a request from a servlet to another resource (servlet, JSP file, or HTML file) on the server.
2. **public void include(ServletRequest request,ServletResponse response)throws ServletException,java.io.IOException:**Includes the content of a resource (servlet, JSP page, or HTML file) in the response.

include() method



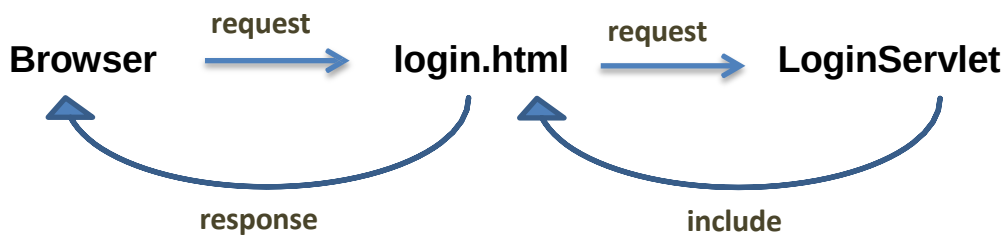
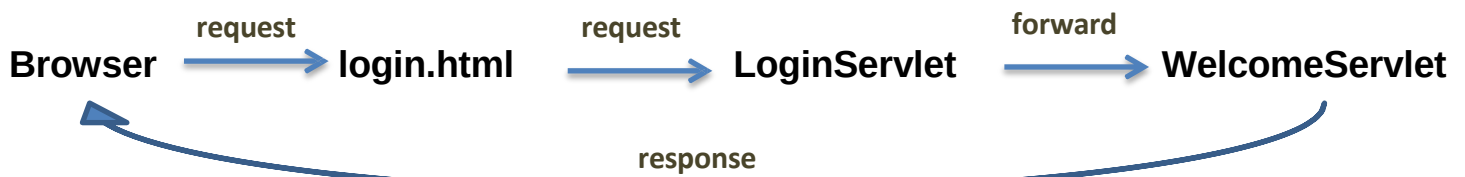
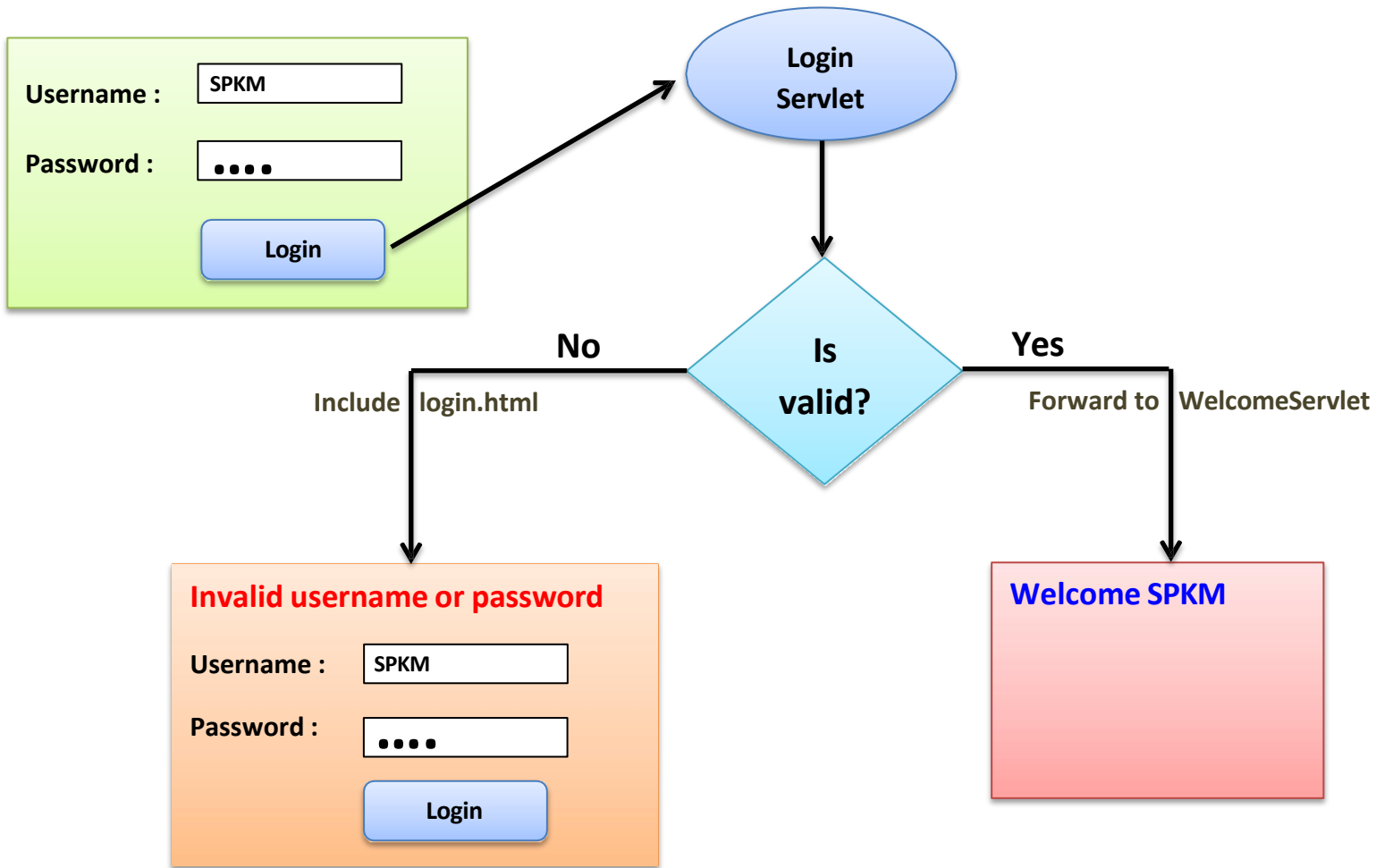
As you can see in the above figure, response of second servlet is included in the response of the first servlet that is being sent to the client.

forward() method:



As you see in the above figure, response of second servlet is sent to the client. Response of the first servlet is not displayed to the user.

Servlet Collaboration Example



- 🚦 **login.html file:** for getting input from the user.
- 🚦 **LoginServlet.java file:** a servlet class for processing the response. If password is valid, it will forward the request to the welcome servlet.
- 🚦 **WelcomeServlet.java file:** a servlet class for displaying the welcome message.
- 🚦 **web.xml file:** a deployment descriptor file that contains the information about the servlet

login.html

```
<html>
  <body>
    <form action="login">
      User Name : <input type="text" name="un"/><br>
      Password : <input type="password" name="pw"/><br>
      <input type="submit" value="Login"/>
    </form>
  </body>
</html>
```

LoginServlet.java

```
import java.io.*;
import jakarta.servlet.*;
import jakarta.servlet.http.*;
public class LoginServlet extends HttpServlet
{
    public void doGet(HttpServletRequest req, HttpServletResponse res) throws
ServletException, IOException
    {
        res.setContentType("text/html");
        PrintWriter out=res.getWriter();
        String user=req.getParameter("un");
        String pass=req.getParameter("pw");
        if(user.equals("spkm")&&pass.equals("123"))
        {
            RequestDispatcher rd=req.getRequestDispatcher("WelcomeServlet ");
            rd.forward(req,res);
        }
        else
        {
            out.print("<font color='red'>Invalid username or password</font>");
            RequestDispatcher rd=req.getRequestDispatcher("login.html");
            rd.include(req,res);
        }
    }
}
```

WelcomeServlet.java

```
import java.io.*;
import jakarta.servlet.*;
import jakarta.servlet.http.*;

public class WelcomeServlet
    extends HttpServlet
{
    public void doGet(HttpServletRequest req,HttpServletResponse res) throws
ServletException,IOException
    {
        res.setContentType("text/html");
        PrintWriter out=res.getWriter();
        String user=req.getParameter("un");
        out.print("<h3><font color='blue'>Welcome "+user);
        out.close();
    }
}
```

login.html

File Edit View History Bookmarks Tools Help

http://localh...81/login.html

localhost:8081/login.html

User Name : spkm

Password : ●●●

Login

valid

File Edit View History Bookmarks Tools Help

http://loca...spkm&pw=123

localhost:8081/login?un=spkm&pw=123

Welcome spkm

invalid

File Edit View History Bookmarks Tools Help

http://local...km&pw=sdfds

localhost:8081/login?un=spkm&pw=sdfds

Invalid username or password

User Name :

Password :

Login